INCH-POUND
MIL-DTL-24646/8C
w/AMENDMENT 1
17 June 2024
SUPERSEDING
MIL-DTL-24646/8C
29 January 2015

DETAIL SPECIFICATION SHEET

FUSE BOXES, FEEDER DISTRIBUTION, SUBMERSIBLE, (15 FEET), 125 VOLTS, ALTERNATING CURRENT, THREE PHASE BUSES, SINGLE PHASE BRANCHES, 0-30 AMPERE MIDGET FUSES SYMBOLS 646.1 AND 647.1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24646.

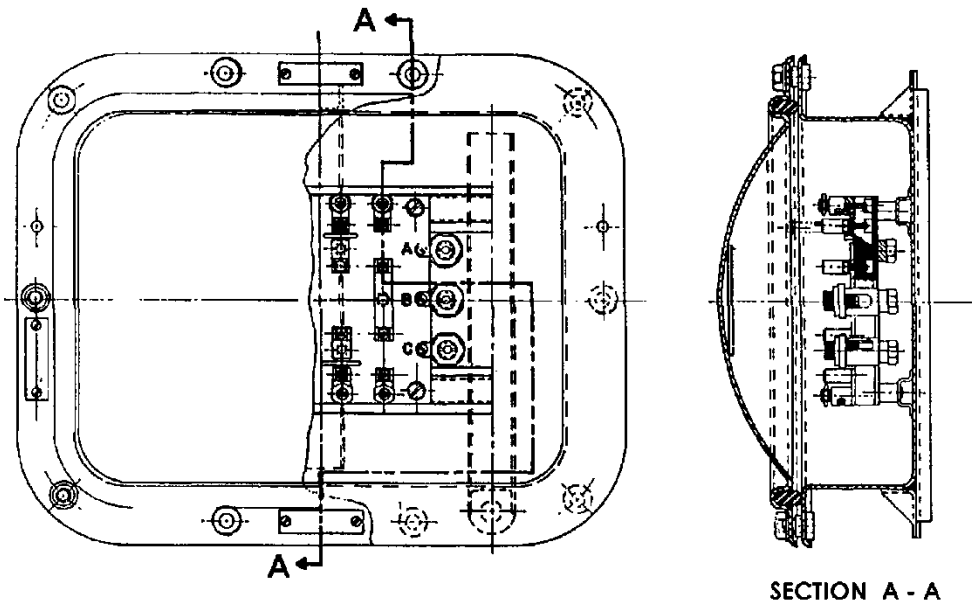
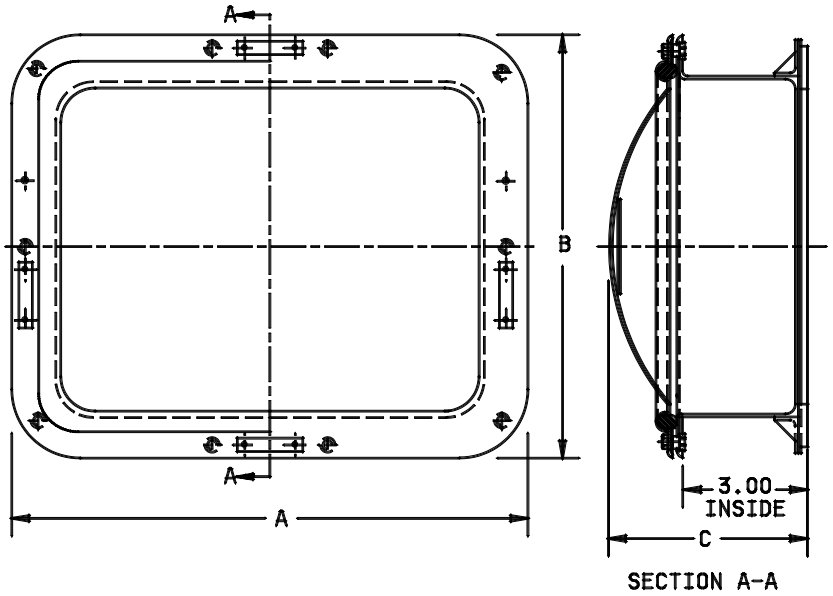


FIGURE 1. Fuse box assembly, typical. Item No. 01, 02
(Symbol 646.1 shown)

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified herein, tolerances shall be as follows:
   Two place decimals - plus or minus 0.02.
   Three place decimals - plus or minus 0.005.

AMSC N/A
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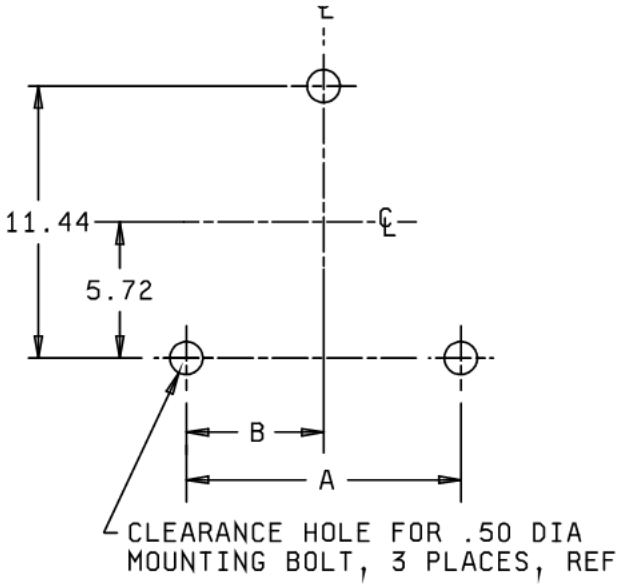
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SYMBOL NO	DIMENSIONS (REF)		
	A	B	C
646.1	16.38	13.38	5.62
647.1	21.38	13.38	5.62

FIGURE 2. Enclosure, overall dimensions. Item No. 03, 04
(Symbols 646.1 and 647.1)

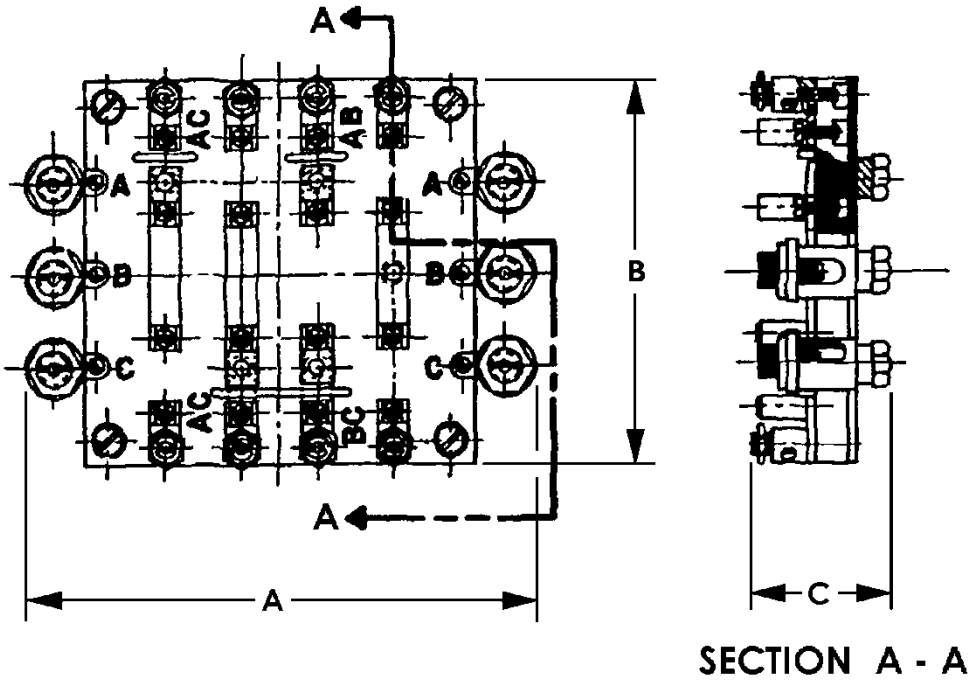
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SYMBOL NO	DIMENSIONS (REF)	
	A	B
646.1	9.25	4.62
647.1	14.25	7.12

FIGURE 3. Mounting dimensions, ref.
(Symbols 646.1 and 647.1)

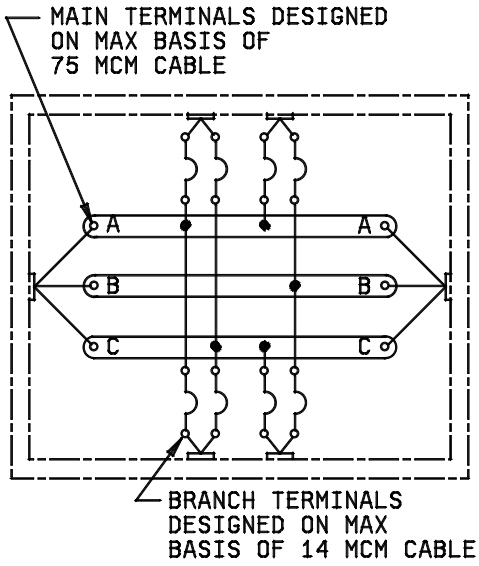
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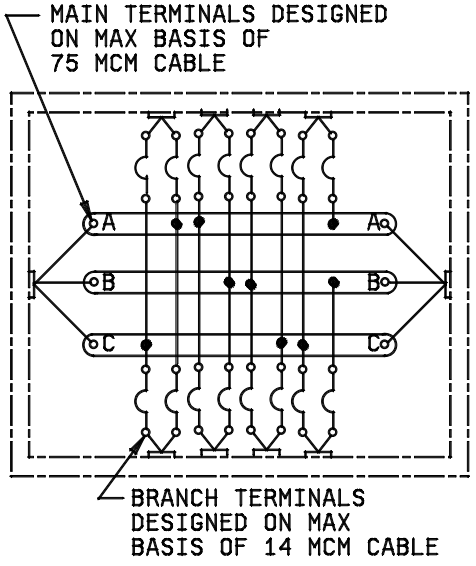
SYMBOL NO	DIMENSIONS (MAX)		
	A	B	C
646.1	7.75	5.75	2.30
647.1	12.63	5.75	2.30

FIGURE 4. Fuse block assembly, typical. Item No. 05, 06
(Symbols 646.1 and 647.1)

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A. Four single-phase branches
(Symbol 646.1)



B. Eight single-phase branches
(Symbol 647.1)

FIGURE 5. Wiring diagrams. Item No. 07, 08

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REQUIREMENTS

Symbol numbers: See table I.

Design:

Electrical rating: 125 volts alternating current, 0-30 ampere, 60 hertz.
Dimensions and configuration: See figures 1, 2, 3 and 4.
Wiring diagrams: See figure 5.
Mounting dimensions: See figure 3.
Degree of enclosure: Submersible (15 feet).
Designation and marking: See MIL-DTL-24646.
Parts: The parts listed in table II and specified in MIL-DTL-24646 shall be used in the construction of the fuse boxes specified herein.

General information:
Military part numbers and general information are as specified in table I.

TABLE I. Military part numbers and general information.

Military part no.	Symbol no.	Approx Weight in pounds 1/	No. of circuits	Feeder MCM	Branch MCM	Fuse size	Bus bar size	Supersedes	
								Drawing no.	Type
M24646/8-001	646.1	27.8	4	75	14	midget 0-30A	0.62 x 0.25	9000-S-74199	0-8A
M24646/8-002	647.1	33.3	8	75	14	midget 0-30A	0.62 x 0.25	9000-S-74202	0-24A

1/ The weight of these fuse boxes may vary + or - 15 percent.

TABLE II. Parts list.

Item no. 1/	Description	QTY M /-		Material/ specification	Remarks
		001	002		
01	Fuse box assembly (sym 646.1)	-		MIL-DTL-24646/8	See figure 1
02	Fuse box assembly (sym 647.1)		-	MIL-DTL-24646/8	See figure 1
2/ 03	Enclosure (10 x 13 x 3)	1		M24142/6-004	See figure 2 Type A Mod
2/ 04	Enclosure (10 x 18 x 3)		1	M24142/6-006	See figure 2 Type A Mod
05	Fuse block assembly	1		MIL-DTL-24646/8	See figure 4
06	Fuse block assembly		1	MIL-DTL-24646/8	See figure 4
07	Wiring diagram	1		MIL-DTL-2036	See figure 5
08	Wiring diagram		1	MIL-DTL-2036	See figure 5
09	Fuse clip (midget 0-30A)	16	32	A-A-55507/1-04	N/A
10	Identification plate	1	1	Brass, ASTM B36/B36M B121/B121M	N/A
11	Information plate	4	8	Brass, ASTM B36/B36M B121/B121M	N/A

1/ Item numbers after the figure titles are the same as the item numbers shown in this table.
2/ Enclosure is same as MIL-E-24142/6, except for inside depth which is 3 inches instead of 5 inches as shown.

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Referenced documents. In addition to MIL-DTL-24646, this document references the following:

MIL-DTL-2036 MIL-E-24142/6 ASTM B36/B36M ASTM B121/B121M A-A-55507/1

The margins of this specification are marked with vertical line to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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(Project 5920-2024-025)

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